Modeling Railways in OSM: from 2D to 1D

State of the Map Europe
Vienna, 17 July 2011

Berend Schotanus
First Flamingo Enterprise B.V.

Overview

- **1. Abstract**
  - needs and expectations regarding OSM
  - translating between semantic systems
- **2. Concrete**
  - creating a heartline

Goal

- Using OSM data for railway purposes
- Route finding
- Passenger information
- Capacity simulation



=> Bringing the data in a format that meets the purpose

Reference Systems

- Postal:
  Stationsplein 2
  3901 ES Veenendaal

- Cartesian:
  lat = ”52.02822”
  lon = ”5.53104”

- Mileage:
  route = ”nl.os05”
  km = ”32.853”

Describing a Railway

-

Levels of detail

- Difference between zoom levels in mapping is huge (19 levels in tile server)
- Desired level depends upon your purpose
- OSM has very fine detail...
- ... but difficult to get lower level of detail

Station level

- Helping passengers find their way
- Requires high level of detail
- Challenge: dealing with multiple physical levels
- Really useful



Track level

- Showing individual tracks and switches
- OSM has incredible level of detail
- Railway traffic control uses mimic diagram, no geographical accuracy

Network level

- Basic function of a railway is moving over the network

- Many questions are network related

- This is really what I'm looking for!



Approach

- Define a heartline that defines the position of a railway
- Strive for geographical accuracy
- Leave out details (no switches etc.)
- Couple to a ‘mileage’ measurement system
- Provide hooks for additional information

Creating the Heartline

Groningen

Leeuwarden

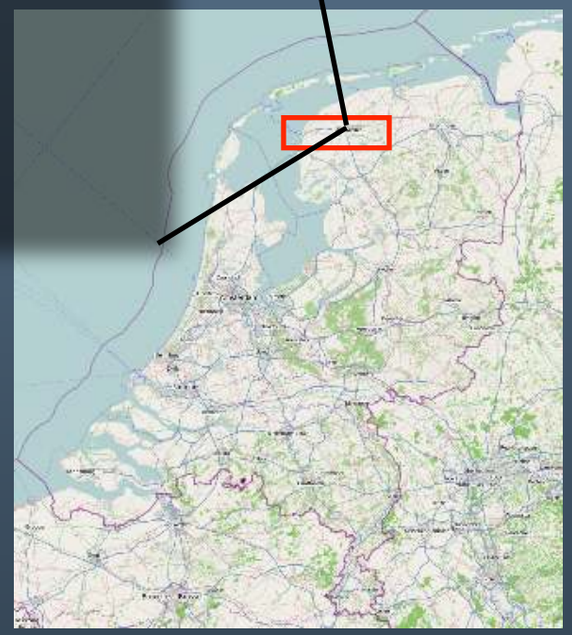
Van Harinxmakanaal
(1951)

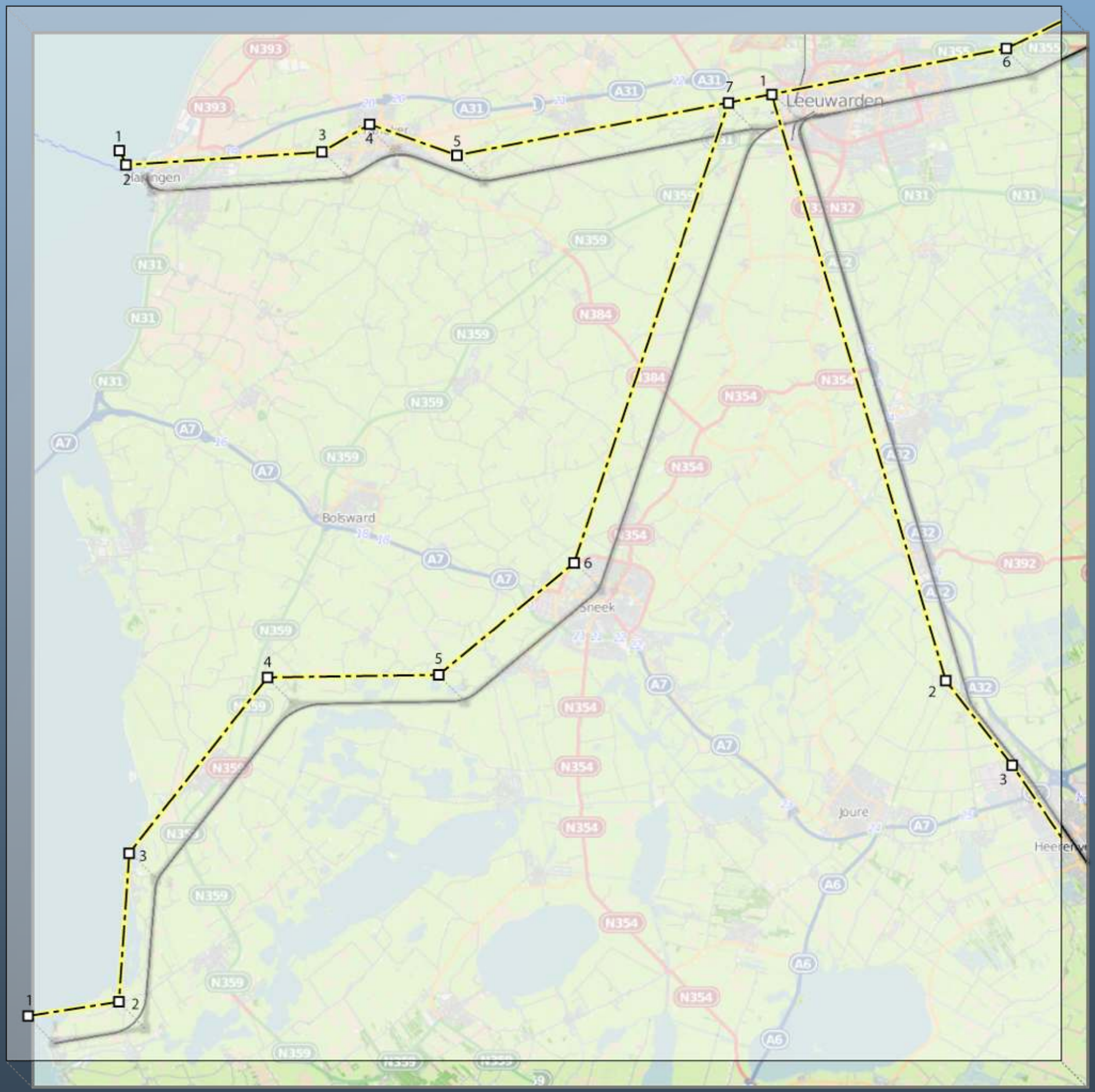
yard

Harlingen

Stavoren

Zwolle




Leeuwarden
Bolsward
Sneek
Joure
1
2
3
4
5
6
7

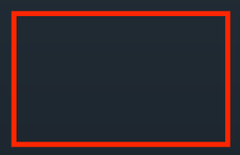
Leeuwarden

Groningen

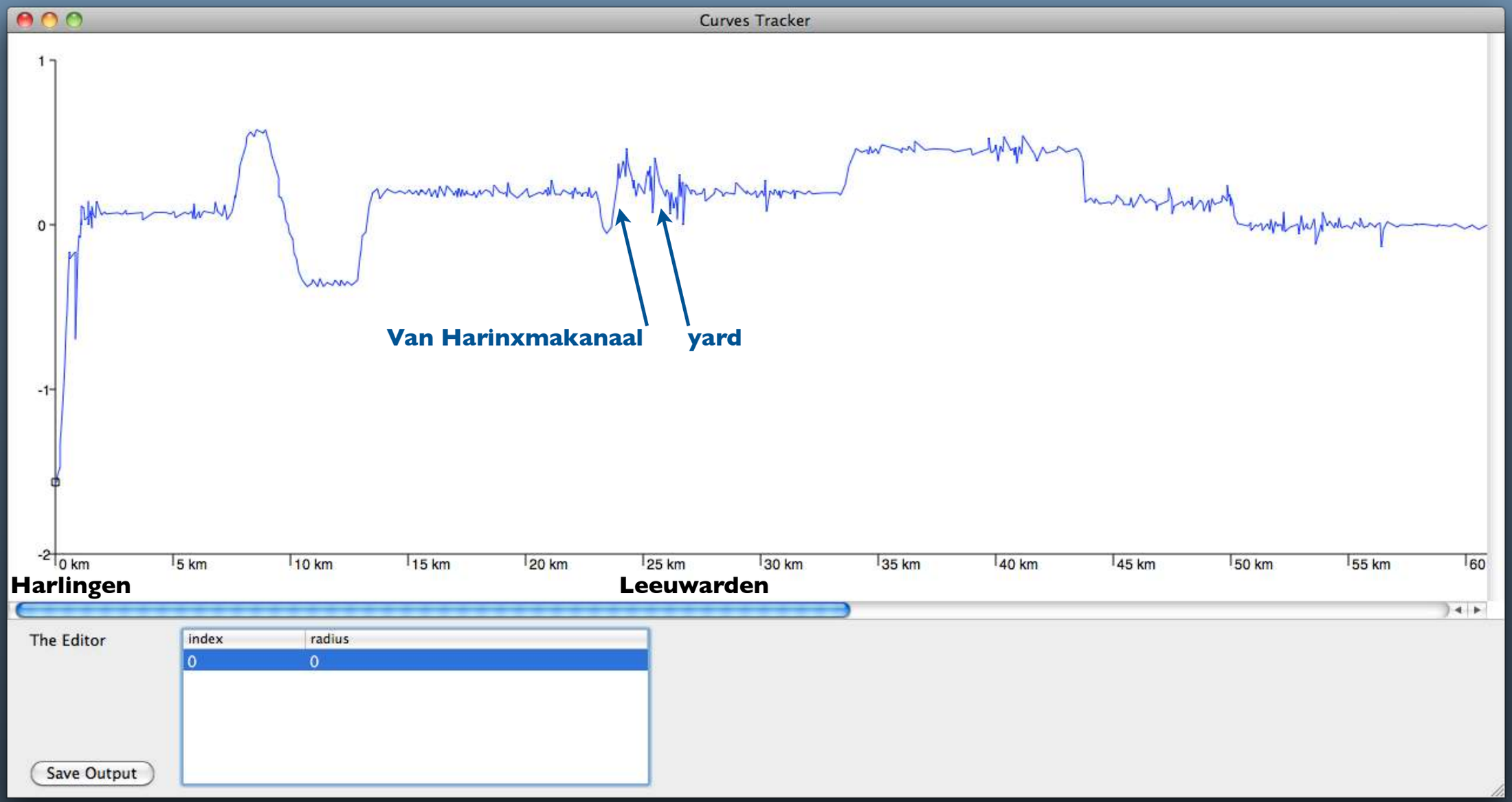
Van Harinxmakanaal

yard

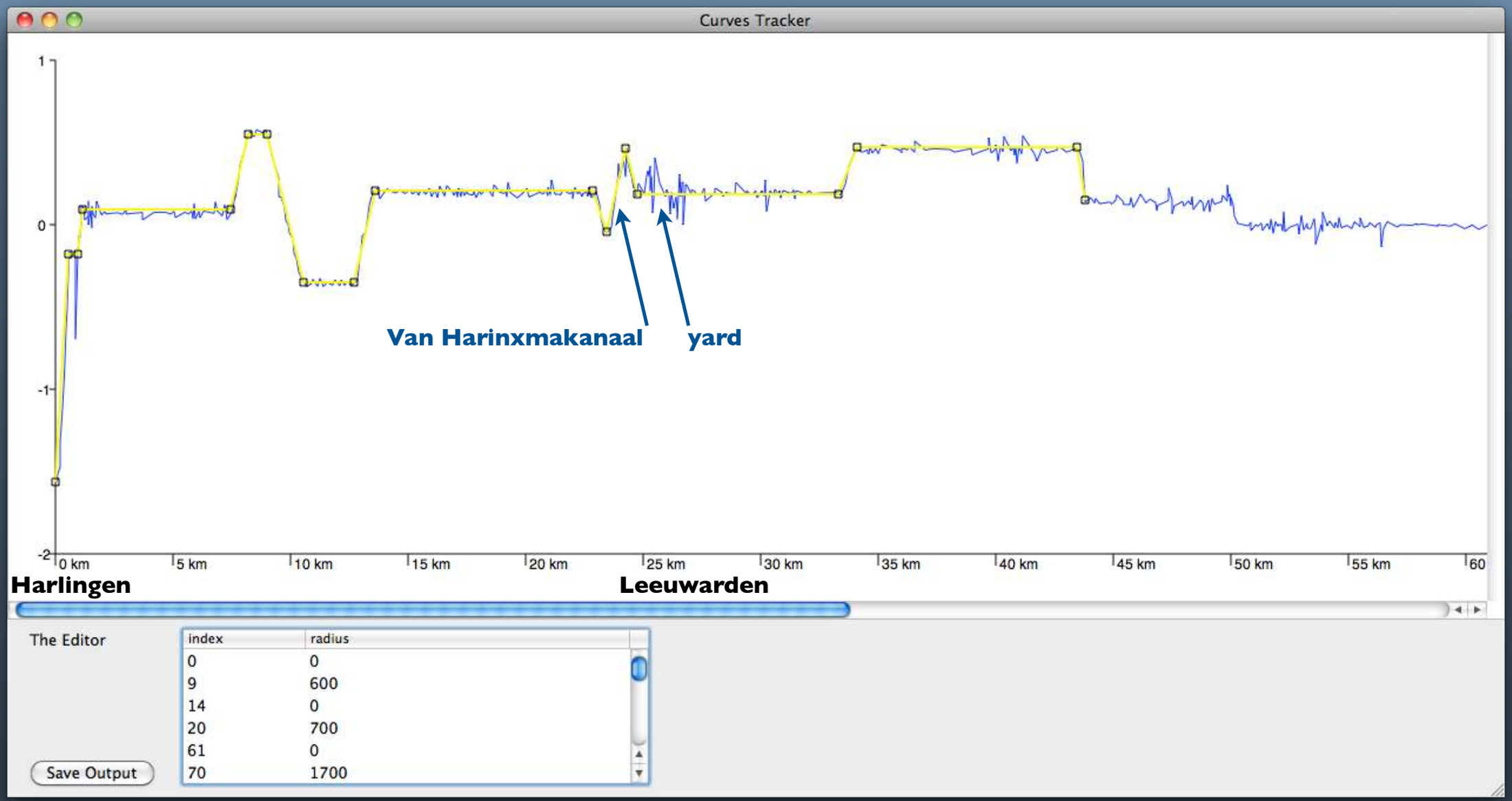
Harlingen



Tracking the curves



Tracking the curves



Leeuwarden

Groningen

Straight line through
the yard

Inaccuracy +/- 50 m

Harlingen

Ignored (small) curve



Leeuwarden

Groningen

Harlingen

Straight line through
the yard

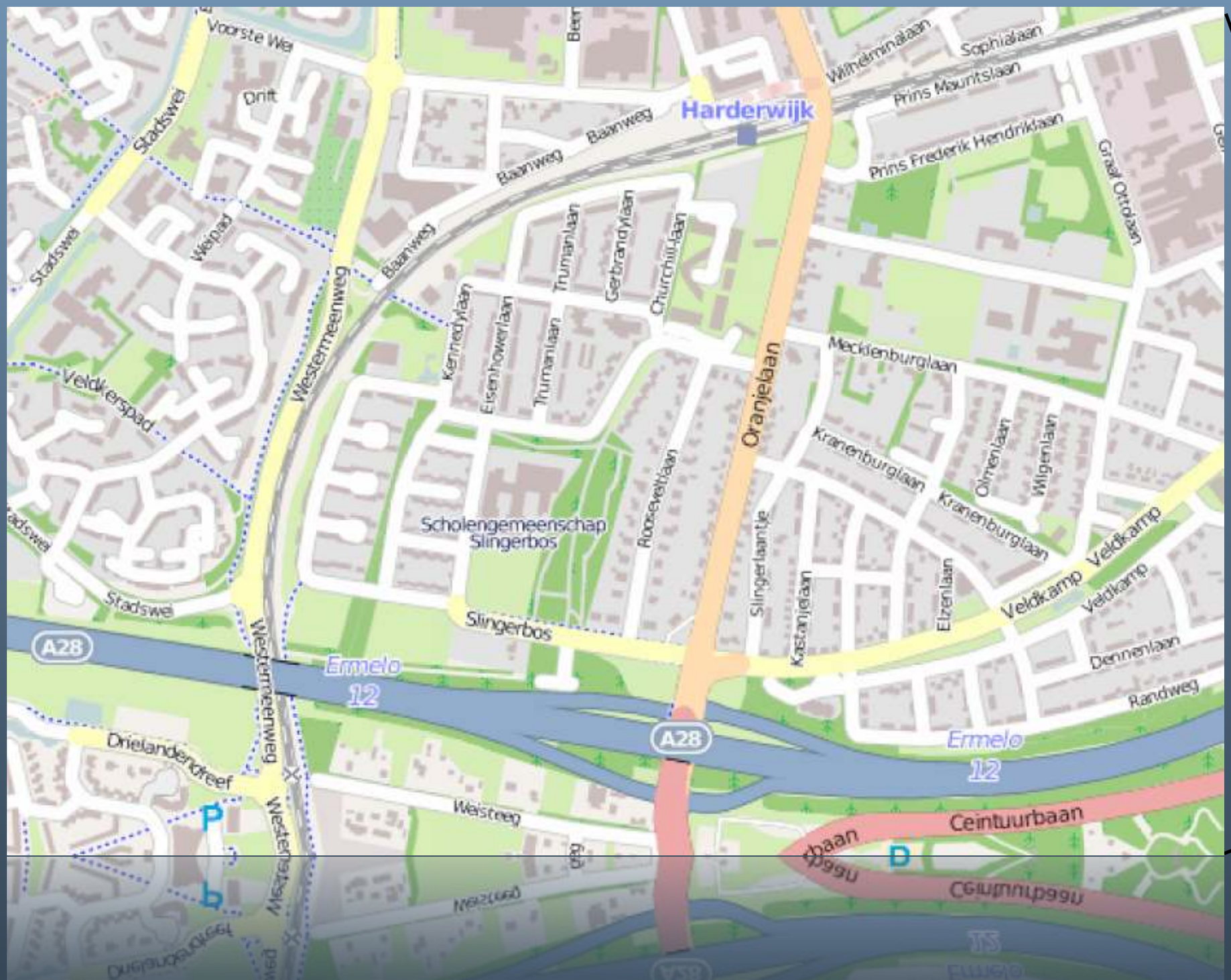
Inaccuracy +/- 50 m

Ignored (small) curve

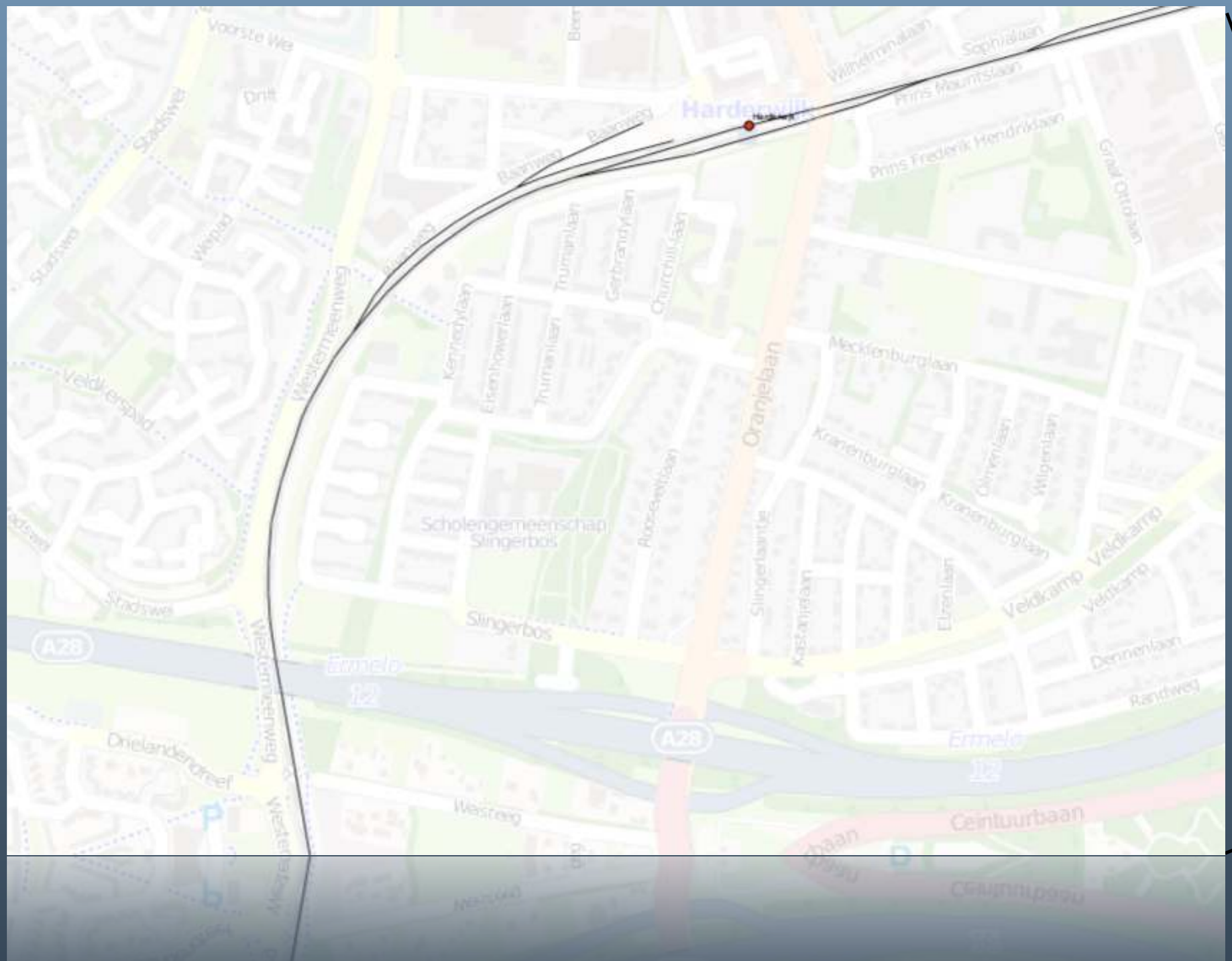
Leeuwarden

Harlingen

Modeling curves

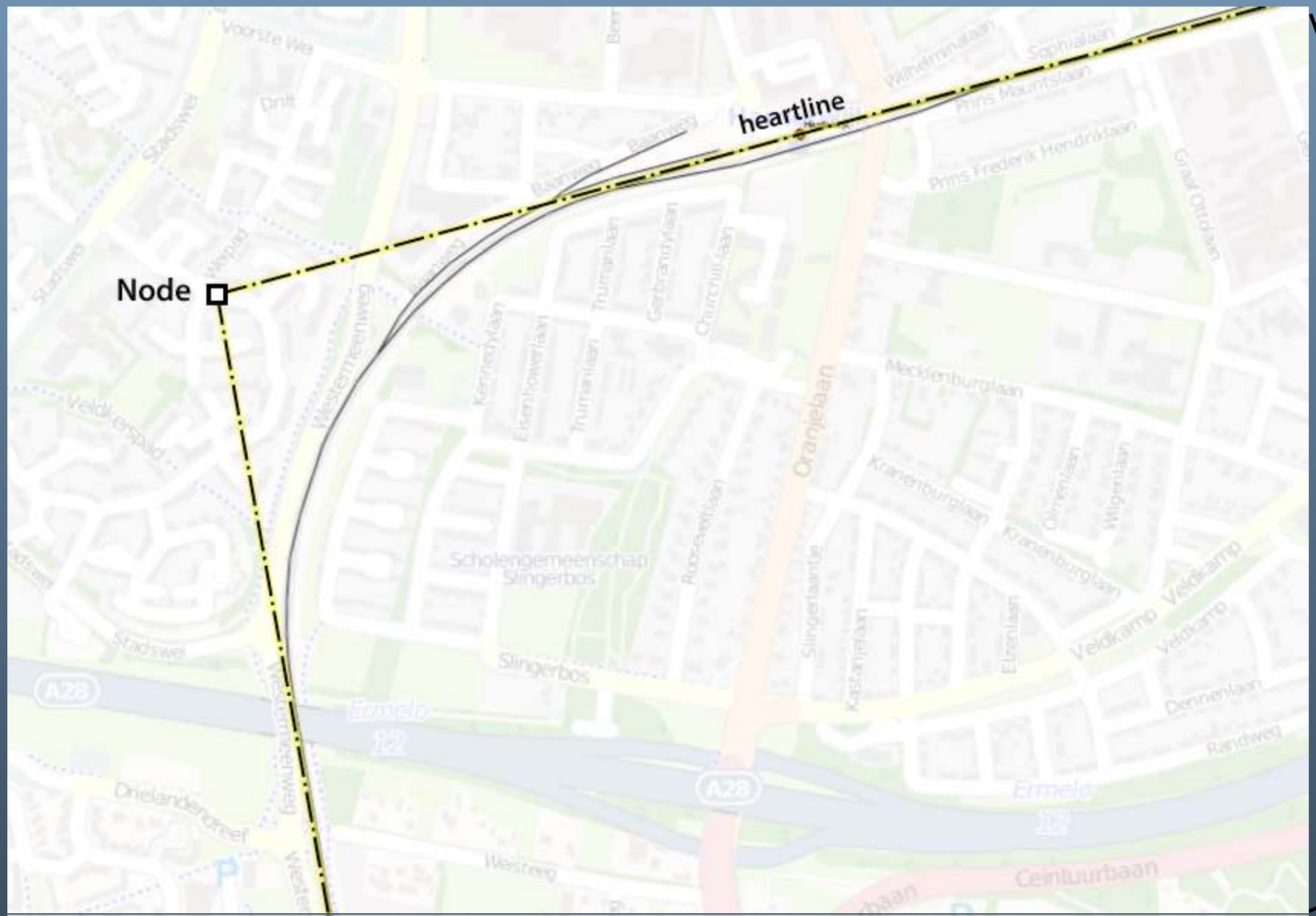


Modeling curves





Modeling curves



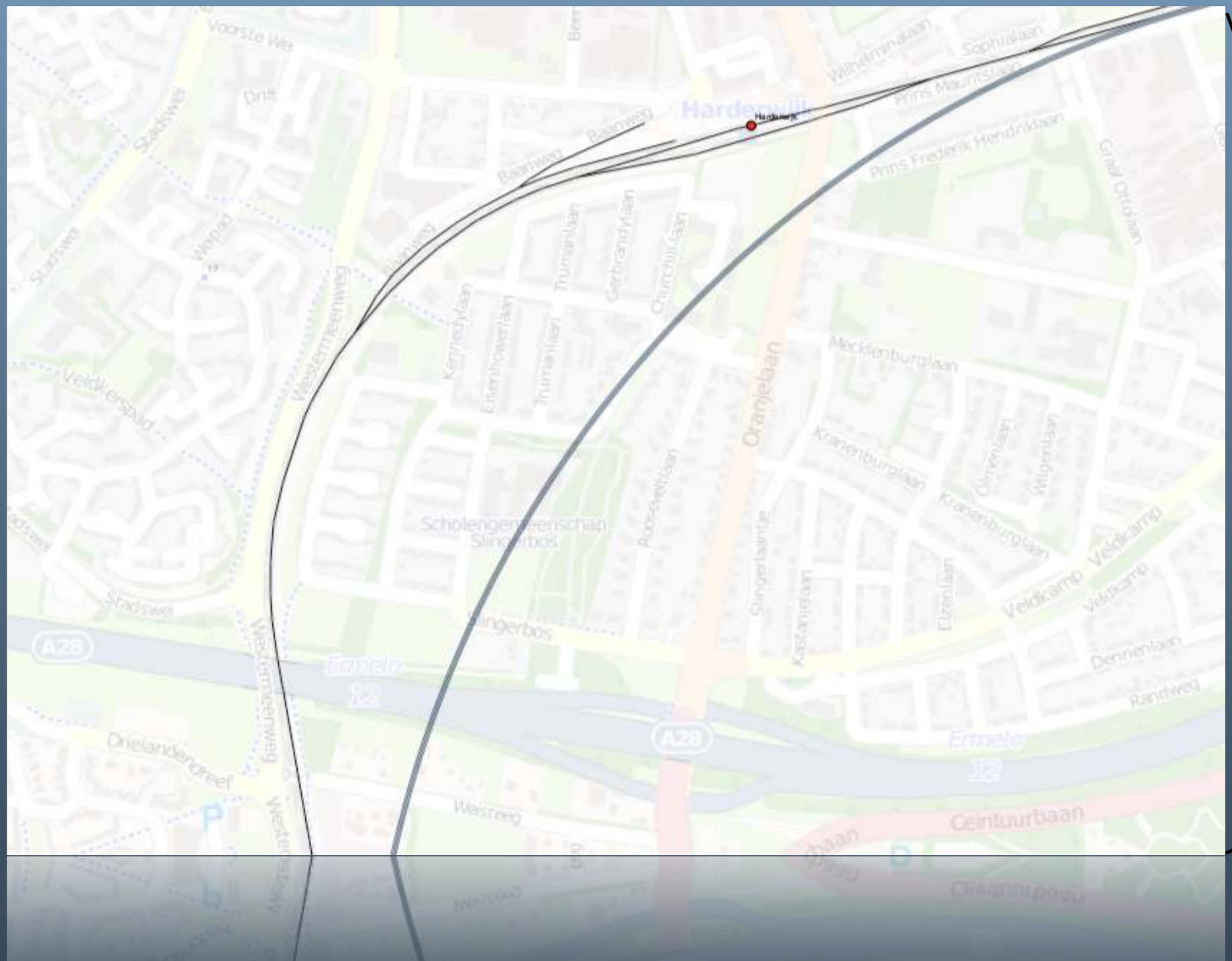


Modeling curves



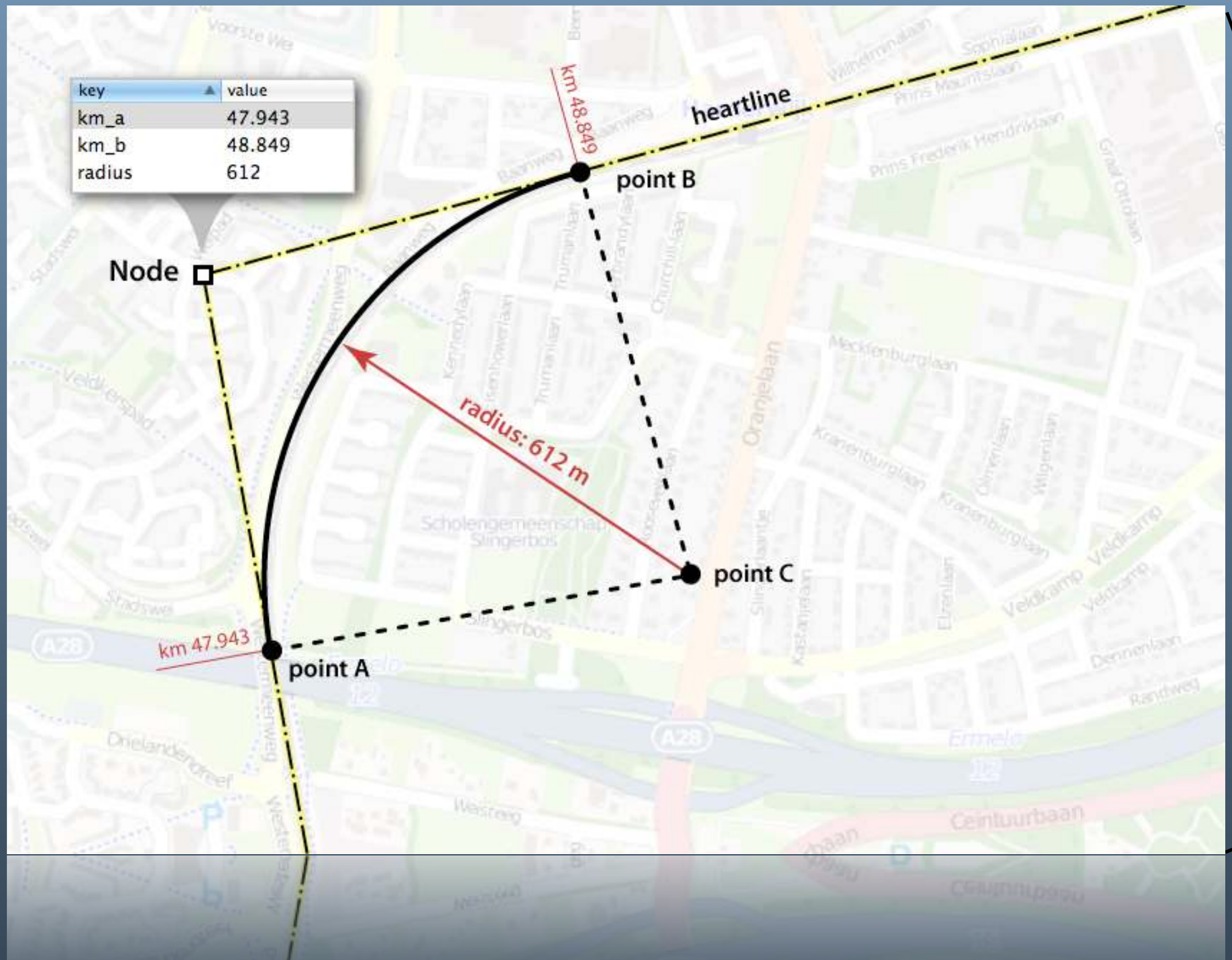


Modeling curves





Modeling curves





Discussion

- It works!
- An editor is needed
- Heartline is useful as a template
- ... but how to share it?
- Could OSM be a source of aggregated data?